

EE 446 HOMEWORK 1 DISCUSSION ANSWERS

Sources checked on May 19, 2026.

Problem 1(e)

Yes. The programming notebook implements an additional compression experiment that combines a smaller student architecture with full integer quantization. The model uses one hidden layer with 16 ReLU units and a softmax output layer. It is trained with an output-based knowledge-distillation loss, then converted to a fully int8 TFLite model.

The main size reduction comes from reducing the number of Dense-layer parameters before quantization. The baseline model has 64-unit and 32-unit hidden layers, while the proposed model has only one 16-unit hidden layer. Full integer quantization then stores the model weights and activations with 8-bit integer tensors instead of float32 tensors. This combination is smaller than using quantization, pruning, or the larger KD student model alone.

The final notebook run reported that the smallest model before Part 1(e) was the baseline int8 model at 5.88 KB with 98.15% test accuracy. The proposed tiny KD int8 model was 2.47 KB with the same 98.15% test accuracy. This is a 57.99% size reduction relative to the previous smallest model, without reducing the measured test accuracy on the held-out split.

Problem 2.1

The current Edge Impulse learning-block documentation lists these built-in learning blocks: Classification (Keras), Regression (Keras), Anomaly Detection (K-means), Anomaly Detection (GMM), Visual Anomaly Detection (FOMO-AD), Image Classification using transfer learning, Keyword Spotting using transfer learning, Object Detection using MobileNetV2 SSD FPN, Object Detection using FOMO, and Classical ML. The Classical ML page currently states that Classical ML is Enterprise-only, so I would not count it as free-tier for a Developer/free account.

Free-tier learning-block categories and examples:

- ? Supervised classification: Classification (Keras), such as motion classification from accelerometer features.
- ? Supervised regression: Regression (Keras), such as predicting a continuous sensor value or weight estimate.
- ? Transfer-learning classification: Image Classification using transfer learning, such as defect/no-defect image classification; Keyword Spotting using transfer learning, such as detecting a wake word.
- ? Object detection: FOMO or MobileNetV2 SSD FPN object detection, such as detecting small objects in camera frames.
- ? Unsupervised or semi-supervised anomaly detection: K-means anomaly detection or GMM anomaly detection, such as flagging abnormal vibration signatures; Visual Anomaly Detection (FOMO-AD), such as detecting visual defects.

? Custom learning blocks: Edge Impulse documentation says custom learning blocks are available for all users, so a user can bring a custom Keras, PyTorch, or scikit-learn training pipeline if the built-in blocks are not enough.

I did not find a general self-supervised learning block in the standard free-tier learning-block list.

Problem 2.2

Under the visual neural-network/classifier layer interface, the documented layer types include:

- ? Dense: A fully connected layer used to learn combinations of the extracted features.
- ? Reshape: Changes tensor shape without changing values, usually to prepare data for another layer type.
- ? Flatten: Converts multi-dimensional feature maps into one vector, commonly before Dense layers.
- ? Dropout: Randomly disables a fraction of units during training to reduce overfitting.
- ? 1D convolution: Extracts local patterns from sequential or time-series style features.
- ? 1D pooling: Reduces the length of 1D feature maps, often by max or average pooling, to lower computation and improve robustness.
- ? 2D convolution: Extracts spatial features from image-like inputs or spectrograms.
- ? 2D pooling: Reduces 2D feature-map dimensions after convolution.
- ? Batch normalization: Normalizes intermediate activations to stabilize and speed up training; it appears in the Edge Impulse Keras visual layer type API.
- ? Output layer: Produces the final task output. For classification this is typically a softmax output that returns class probabilities.

Expert mode exposes the broader TensorFlow/Keras API, but the list above is the practical visual-layer set relevant to the normal Classifier architecture UI.

Problem 2.3

The current Deployment documentation lists these model-compression or optimization options:

- ? Model version: quantized int8 model versus unoptimized float32 model.
- ? Compiler option: standard TFLite/LiteRT versus EON Compiler.
- ? EON Compiler RAM optimized: an additional RAM-saving mode when available. The pricing page states that the Developer/free plan excludes RAM Optimized mode, so this should be treated as Enterprise-only rather than free-tier.

For a free-tier account, the practically available compression choices are int8 quantization and the EON Compiler for models trained in Edge Impulse. The unoptimized float32 model is the baseline comparison rather than a compression method.

Problem 2.4

The Synthetic data documentation lists built-in blocks for DALL-E image generation, FLUX Pro image generation, Time-series Data Augmentation, and Whisper Keyword Spotting Generation. The same documentation marks Time-series Data Augmentation as Enterprise-only, so the free-tier modalities I would report are:

? Image: DALL-E image generation and FLUX Pro image generation. These generate image samples from text prompts. The DALL-E block uses an OpenAI API key, and the FLUX Pro block uses Replicate.

? Audio, voice, or keyword spotting: Whisper Keyword Spotting Generation. This generates speech or keyword audio from text/keyword prompts using an OpenAI API key.

Time-series data augmentation is documented, but it is Enterprise-only, so I would not list it as a free-tier method. Custom synthetic-data blocks are also Enterprise-only.

Synthetic data is important because it can fill gaps in real datasets, create examples for rare or expensive-to-collect cases, improve robustness through augmentation, and reduce the time and cost required to collect enough labeled data. It is especially useful in edge ML because device data can be limited, imbalanced, privacy-sensitive, or difficult to collect under every operating condition.

Sources

? Edge Impulse Learning blocks: <https://docs.edgeimpulse.com/studio/projects/learning-blocks>

? Edge Impulse Classical ML: <https://docs.edgeimpulse.com/studio/projects/learning-blocks/blocks/classical-ml>

? Edge Impulse neural-network layers: <https://docs.edgeimpulse.com/knowledge/concepts/machine-learning/neural-networks/layers>

? Edge Impulse Keras visual layer type API: https://docs.edgeimpulse.com/tools/libraries/api-bindings/studio/python/edgeimpulse_api/models/keras_visual_layer_type

? Edge Impulse Deployment: <https://docs.edgeimpulse.com/studio/projects/deployment>

? Edge Impulse EON Compiler: <https://docs.edgeimpulse.com/studio/projects/deployment/eon-compiler>

? Edge Impulse Synthetic data: <https://docs.edgeimpulse.com/studio/projects/data-acquisition/synthetic-data>

? Edge Impulse Pricing: <https://www.edgeimpulse.com/pricing>